Database data types and validation

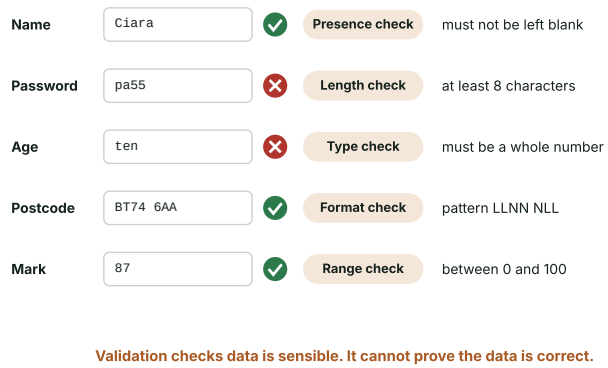
L09 FACT FILE

Specification: students should be able to

- Identify and use appropriate data types when creating a database structure
- Justify the choice of data type for a field
- Demonstrate understanding of the need for data validation
- Describe the following types of validation checks: presence, length, type, format and range
- Write simple validation rules for database fields

CCEA fact file: Fact File 08: Database Applications (2)

Key words: data type, Short Text, Long Text, Number, Currency, Date/Time, AutoNumber, Yes/No, OLE Object, Hyperlink, Attachment, Calculated, Lookup Wizard, validation, presence check, length check, type check, format check, range check, input mask, validation rule, validation text, verification



Five validation checks applied to form fields, with passes and failures

Data types

When a database table is designed, every field is given a field name and a data type. The data type decides what kind of data the field can hold and what can be done with it. Short Text is for names, addresses, postcodes and any numbers that are not used in calculations, such as phone numbers. Long Text holds longer descriptions. Number is for numeric values that will be used in calculations, such as the number of people in a group. Currency is for money and shows two decimal places. Date/Time stores dates and times so that they can be sorted in date order and used to work out ages or due dates. AutoNumber creates a new number automatically for each record and is often used for the key field. Yes/No holds a Boolean value with only two options. Choosing the right data type is the first step in keeping data accurate, because the database will refuse data of the wrong type.

More data types and justifying your choice

Database packages such as Access offer several more data types. Hyperlink stores a web address or email address that can be clicked. Attachment lets files, such as a photo of a club member, be attached to a record. OLE Object, short for Object Linking and Embedding, stores a file created in another program, such as a spreadsheet or a word-processed document. Calculated holds the result of a calculation that uses other fields in the same record, for example a total cost worked out from the number in the group multiplied by the price per person. Lookup Wizard creates a drop-down list of values, so the user picks an answer rather than typing it. In the exam you may need to justify a data type. Give a reason linked to the data. For example, a phone number should be Short Text because it starts with a zero, may contain spaces and is never used in a calculation. Storing it as a Number would drop the leading zero.

Why validation is needed

Validation is an automatic check carried out by the computer when data is entered. It makes sure that the data is sensible, reasonable and within acceptable limits. If the data breaks a rule, it is rejected and an error message, called validation text, tells the user what is wrong. Validation matters because a database is only useful if its data is accurate. Wrong or missing data leads to wrong results, for example a letter sent to an incomplete address or a bill calculated from an impossible number of days. Validation reduces these input errors. However, validation cannot check that data is correct. It can only check that data is allowed. If a parent's date of birth is typed as 12/05/1984 instead of 21/05/1984, both dates are valid, so the mistake gets through. This is why validation is often combined with verification, such as typing an email address twice, or a visual check against the original document.

The five validation checks

A presence check makes sure a field is not left blank. It suits fields that every record must have, such as a customer's surname or postcode. A length check makes sure the data has the right number of characters. It can set an exact length, such as a seven-character camper code, or a minimum or maximum, such as a password of at least eight characters. A type check makes sure the data is of the correct data type, so letters cannot be typed into a Number or Date/Time field. A format check makes sure the data matches a set pattern, such as a Northern Ireland postcode that starts with BT followed by digits, a space, a digit and two letters. A range check makes sure a value lies between a lower limit and an upper limit, for example from 1 to 5 days at a summer camp or a test mark from 0 to 100. When answering, always name the check and then describe it using the field in the question.

Writing validation rules in Access

In Access, most checks are set in the field properties in Design View. For a presence check, set Required to Yes, or enter the validation rule Is Not Null. For a length check, set the Field Size, or write a rule using the Len function, such as Len([Password])>=8, which counts the characters in the field. A type check happens automatically once the data type is chosen. For a format check, use an input mask. In a mask, L means a letter must be typed, 0 means a digit must be typed and 9 means a digit is optional, so the mask "BT"09 0LL fits Northern Ireland postcodes. For a range check, write a rule with comparison operators, such as >=1 And <=5, or use Between 1 And 5, which includes both end values. Always add validation text that explains the rule in plain English, so the user knows how to fix the error rather than just seeing that it was rejected.

T Key terms from the mark schemes

Data type
The kind of data a field can hold, e.g. Short Text, Number, Currency or Date/Time. It controls what can be entered and what can be done with the data.
Credited: U1 2019 Q3, 2024 Q3, 2025 Q5

Validation
An automatic check made by the computer as data is input, to make sure the data is sensible, reasonable and inside allowed limits. Data that breaks the rules is rejected; validation cannot confirm the data is correct.
Credited: U1 2021 Q3, 2025 Q5; U4 2025 Q6

Presence check
A check that makes sure data has been entered in a required field, so the field cannot be left blank.
Credited: U1 2021 Q2, 2022 Q10, 2025 Q5

Length check
A check that makes sure the data entered has an exact number of characters, or is no shorter or longer than set limits.
Credited: U1 2022 Q10, 2025 Q5

Type check
A check that makes sure the data entered is of the correct data type, e.g. numeric, text or date.

Format check
A check that makes sure the data entered matches a set pattern, e.g. a postcode or an email address containing @.
Credited: U1 2023 Q3, 2025 Q5

Range check
A check that makes sure the data entered lies between a lower limit and an upper limit.
Credited: U1 2021 Q2, 2022 Q10, 2025 Q5

Verification
A check that data has been entered or copied accurately, e.g. by double entry or a visual check against the source.

Currency
A data type for money values, stored and displayed with two decimal places.

OLE Object
Object Linking and Embedding: a data type for a file or object created by software other than the database, e.g. a spreadsheet or word-processed document.

Hyperlink
A data type that stores a link to a web address (URL) or an email address.

Lookup Wizard
Creates a list of values so that a value can be chosen from a drop-down list and entered into the field.

Input mask
A pattern set on a field that controls which characters can be typed in each position, used to carry out a format check.

AutoNumber
A data type, often used for a key field, in which the computer creates a new sequential number automatically for each record.

Yes/No
A data type that holds one of only two values (a Boolean), such as Yes/No or True/False.

Attachment
A data type that lets files, such as a digital image, be attached to a record.

Calculated field
A data type that stores the result of a calculation using other fields in the same record.

Exam focus

Data types and validation have come up often: 2019 Q3 and 2024 Q3 asked for data types with a justification, 2022 Q10 covered field types and validation, 2021 Q2(g) asked about presence and range checks, 2023 Q3 asked for validation rules, and 2025 Q5 covered data types and validation checks. Most marks come in pairs: one mark for naming the check or data type and one for a reason or description in context. 'Range check' alone will rarely get both marks; 'a range check so that the number of days booked must be from 1 to 5' will. In 2025 many candidates did not realise that validation cannot tell whether data is correct, so learn that point. When writing a rule, check the boundaries carefully: for 5 to 20 people, >=5 And <=20 is right, while >5 And <20 wrongly rejects 5 and 20. Do not confuse a length check, which counts characters, with a range check, which compares values.

Unit 1 2025 Q5	[11]	Data types, validation checks, data processing
Unit 1 2023 Q3	[9]	Database concepts; validation rules
Unit 1 2022 Q10	[11]	Database tools, field types, importing csv, validation
Unit 1 2021 Q2(g)	[3]	Presence and range checks

? Check yourself

1. Why is validation used when data is entered into a database? [2]
2. Explain why validation cannot guarantee that data is correct. Use an example. [2]
3. Describe a range check and give an example of a field where one would be used. [2]

Answers: 1. Any two from: to check data is sensible/reasonable/within acceptable limits (1) / to reduce input errors (1) / so that invalid data is rejected at the input stage (1) / so the database is more accurate/reliable (1). 2. Validation only checks that data is allowed/follows the rules, not that it is the true value (1). Example, e.g. a date of birth typed as 12/05/1984 instead of 21/05/1984 is still a valid date and is accepted (1). 3. Checks that a value lies between a lower and an upper limit (1). Example, e.g. an exam mark must be from 0 to 100 / age 11 to 19 (1).